Rohan W

Senior Machine Learning Engineer | LLM Systems, Inference Optimization, Evaluation Pipelines, GenAI Infrastructure

Bengaluru, India

+91 9840571093 · rohan.w.charles@gmail.com · linkedin.com · github.com

Summary

Senior Machine Learning Engineer with experience building production GenAI platforms, OpenAI-compatible model APIs, evaluation workflows, GPU-optimized inference pipelines, and scalable ML infrastructure. Strong fit for ML Systems, LLM Inference, AI Platform, Applied ML, and Research Engineering roles at teams building foundation-model products, real-time AI systems, and developer APIs.

Technical Skills

- **LLM Systems & GenAI**: OpenAI-compatible APIs, RAG, multi-agent systems, prompt evaluation, fine-tuning workflows, LLM-as-judge, multi-cloud model gateways
- **Inference & Optimization**: vLLM, SGLang concepts, CUDA, CUDA Graphs, TensorRT, ONNX Runtime, quantization, dynamic batching, latency/throughput optimization
- **ML & Modeling**: PyTorch, TensorFlow, BERT, multi-label classification, threshold tuning, class imbalance handling, model evaluation, A/B testing
- **Data & Evaluation**: annotation workflows, synthetic data validation, automated quality checks, error analysis, experiment design, production monitoring
- **Infrastructure & MLOps**: Temporal, Celery, Docker, Kubernetes, Helm, CI/CD, distributed workflows, multi-cloud architecture

Experience

Senior Machine Learning Engineer — **March 2025 – Present**
Avalara — *Bengaluru, IN (Remote)*

- Built and deployed a secure multi-cloud LLM platform exposing standardized **OpenAI-compatible APIs** across **Azure, AWS Bedrock, and Google Vertex AI**, handling **1M+ daily requests** for internal agent and GenAI teams.
- Designed model-routing, governance, and provider-abstraction layers to support production GenAI applications while reducing duplicate integration effort across teams.
- Built a **BERT-based HS-code multi-label classification** system with label design, validation workflows, threshold tuning, class-imbalance handling, and production evaluation.
- Implemented GPU inference improvements using **TensorRT, ONNX, dynamic batching, and efficient scheduling** to improve latency, throughput, and GPU utilization for classification workloads.
- Built the **CPT training pipeline for Qwen 3.5-4B (full fine-tune, DeepSpeed ZeRO Stage 3, 8× A100-40GB)** on 1.58M Avalara tax-code examples; reduced eval cross-entropy 3.2× (0.41→0.13) in one epoch, producing the domain base for downstream QLoRA SFT and reranking

ML Research Engineer - II (L5) — **April 2024 – March 2025**
Amazon — *Bengaluru, IN*

- Improved Items Glance View by **20% across 100M products** through A/B testing, LLM-assisted experimentation, and production deployment for eliminating products defects.
- Built a production system fine-tuning **Mistral 7B with LoRA and HuggingFace**, serving via **vLLM** for catalog enrichment, validated against a Claude-based POC.
- Developed an LLM platform using **Claude 3.5 Haiku** to accelerate Data Associate workflows for missing-attribute generation and catalog backfilling.
- Developed and presented an automated prompt-fine-tuning POC using **DSPy**, projected to reduce manual prompt-engineering effort and improve ML engineer productivity by **140%**; recognized with **RISE Q4 2024**.
- Enhanced Amazon retail search workflows using Chain-of-Thought prompting, meta-prompting, controlled experiments, and production evaluation across internal services.

Senior Machine Learning Engineer — **Feb 2020 – June 2024**
Rapid Acceleration Partners — *Chennai, IN*

- Led and mentored a **4-member ML team** building document AI, OCR, classification, chatbot, and workflow-automation systems for production customers.
- Re-architected parts of the **RAPFlow** ML pipeline for more efficient GPU deployment, adding **GPU-level inference optimizations**, runtime/export improvements, and batch-processing changes.
- Optimized PaddleOCR invoice pipeline by writing custom **CUDA kernels** for GPU-side image normalization and **HWC to CHW preprocessing, and DBNet** probability map binarization post-processing, combined with **TensorRT FP16** engine compilation
- Built GPU-optimized document classification and document-understanding pipelines for the airplane-parts domain using **TensorRT, ONNX, and batching** to improve latency and throughput.
- Developed a GPT-3.5 and LangChain chatbot for automotive dealerships to answer customer questions, provide product information, schedule appointments, reduce drop-off, and improve the sales funnel.
- Led delivery of AI automation tools for US car dealerships and warehouse workflows, improving lead generation, operational throughput, and customer workflow efficiency.

Data Science Associate — **Dec 2018 – Nov 2019**
ZoomRx — *Chennai, IN*

- Built data-processing tools to integrate data from multiple sources into downstream analytics and operational modules, improving reliability of recurring data workflows.

Education

SRM University — **July 2015 – July 2019**
B.Tech, Electronics and Communication Engineering — *Chennai, TN*

OSS Contributions

- Active contributor to open-source ML and systems libraries, with merged pull requests across core Python runtime, LLM infrastructure tooling, and optimization frameworks. Full contribution history available on GitHub.

Certifications

Deep Learning Specialization — **Nov 2019**
Coursera

Game Theory — **Sept 2020**
Stanford University